Astana IT University
School of Software Engineering

Diploma Project

Development of a System for Detecting and Anonymizing Personal Data Before Sending Text to Artificial Intelligence Services

User Interface Prototype

Authors: Dossan Amina, Abzhamal Nazerke
Educational programme: Software Engineering
Supervisor: Sakhipov A.A., PhD,
Assistant Professor

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1. Introduction

This document presents the user interface (UX) prototype of the PII Anonymization Gateway. The prototype shows how users interact with the browser extension when sending text to ChatGPT or Gemini, how administrators configure the system, and how internal systems use the programming interface. Its purpose is to validate the requirements defined in the document *Requirements Specification* before implementation and to serve as the basis for user testing.

2. Prototype Description

The prototype is a high-fidelity static prototype consisting of 18 screens. The main user scenario (screens U1–U4) is clickable and can be walked through in order. The interface of the user screens is in Russian, screen U6 shows the Kazakh interface, and the extension supports Kazakh, Russian and English (FR-5.7). All names, identifiers, phone numbers and addresses on the screens are fictitious; IIN, BIN, IBAN and card numbers are generated in a valid format.

The screens are grouped as follows:

- *User scenario in the browser extension* (U1–U8): entering text, reviewing detected data, preview of the anonymized text, restored response, blocking by policy, Kazakh and mixed-language text, and failure of the anonymization server;
- *Extension window* (X1, X2): sign-in and status;
- *Administration panel* (A1–A6): sign-in, statistics, policies, log, users and roles, security settings;
- *Programming interface* (P1): examples of API requests and responses.

The exported screens are stored in the `docs/ux-prototype/screens` directory of the repository.

3. Design Decisions Based on the User Survey

The prototype reflects the results of the user survey ($n = 35$) described in the document *Evidence of Relevance* (Table 1).

Table 1: Survey findings and design decisions

Finding	Design decision	Screens
57% want to review detected data before sending; 14% prefer automatic replacement	Review before sending is the default mode; the automatic mode can be enabled in the extension window if the administrator allows it	U2, X1
51% prefer detecting all personal data even with occasional false alarms	Low-confidence detections are shown as unmarked suggestions; the user can mark missed data manually	U2, U7
71% use several languages or mix them; 77% use Kazakh or mixed prompts	Detected languages are shown; the extension interface is available in Kazakh	U6, U7
89% accept an additional delay of at least 1–3 s	Processing time is shown to the user and to the administrator	U3, A2
Concerns about confidentiality and trust in the tool	The user sees exactly what the AI service receives; the log contains no original values; text is not sent if it could not be checked	U3, A4, U8
54% want a detection log and per-type rules	Policies per data type and a filterable log	A3, A4

4. Main User Scenario

The main scenario corresponds to use cases UC-1 and UC-2. The user writes a request in ChatGPT that contains the name, IIN, identity document number and IBAN of a client, the BIN of an organization, an address and the contacts of a manager.

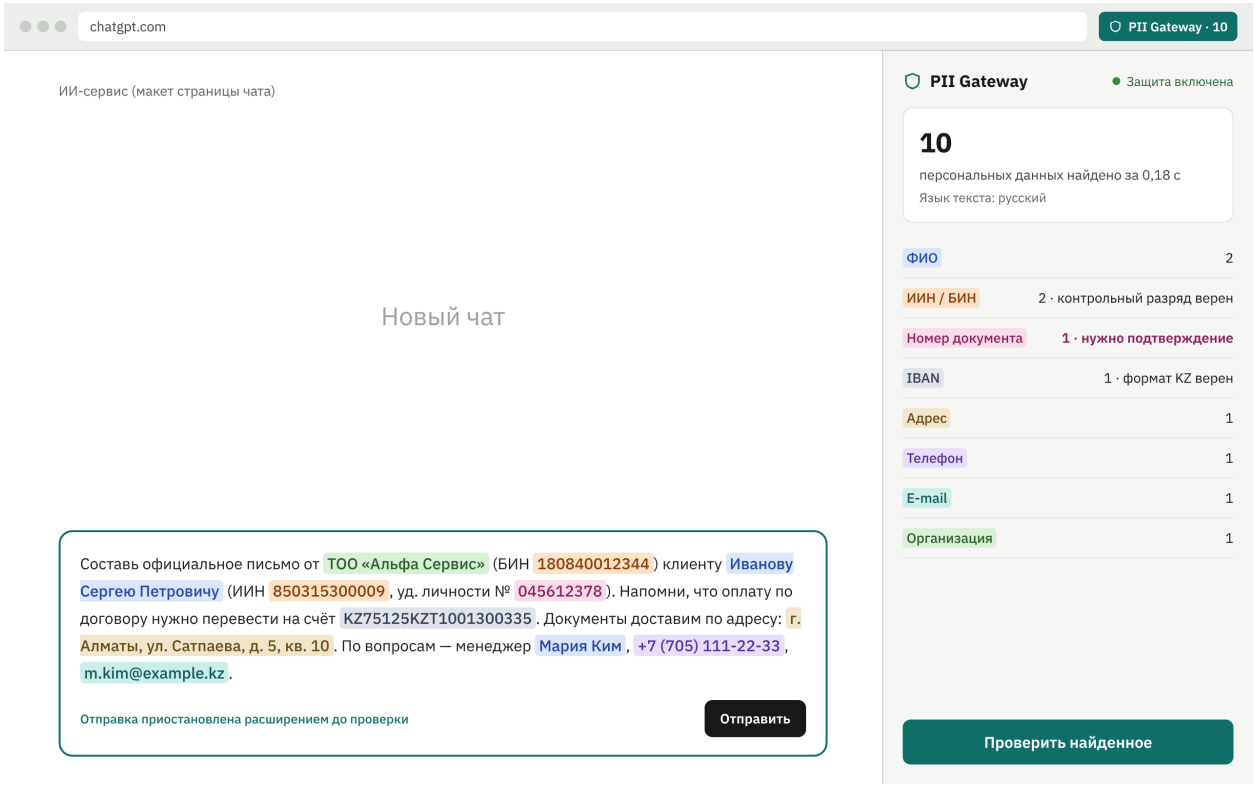


Figure 1: U1. Text input: detected personal data are highlighted by type directly in the input field; IIN and BIN are marked as validated by the check digit

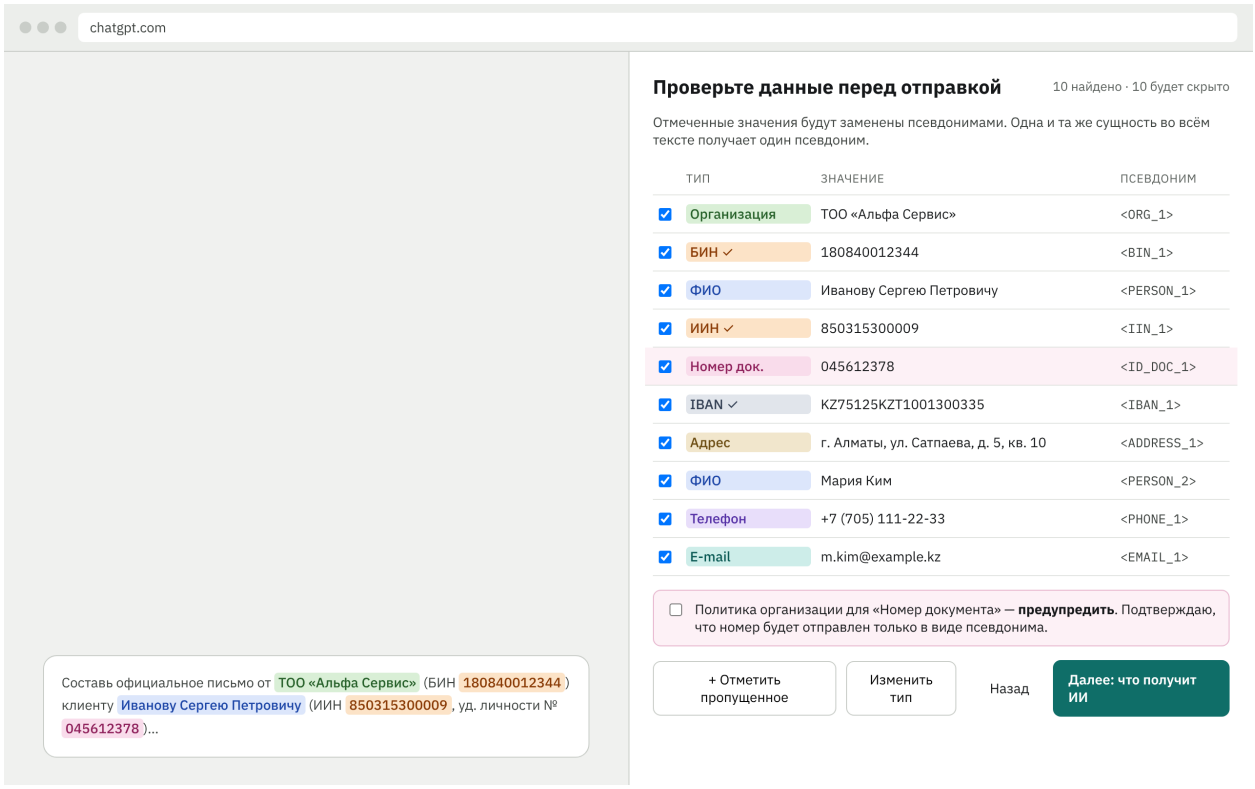


Figure 2: U2. Review of detected data: the user can unmark a value, change its type or mark missed data; the policy “warn” for the identity document number requires explicit confirmation

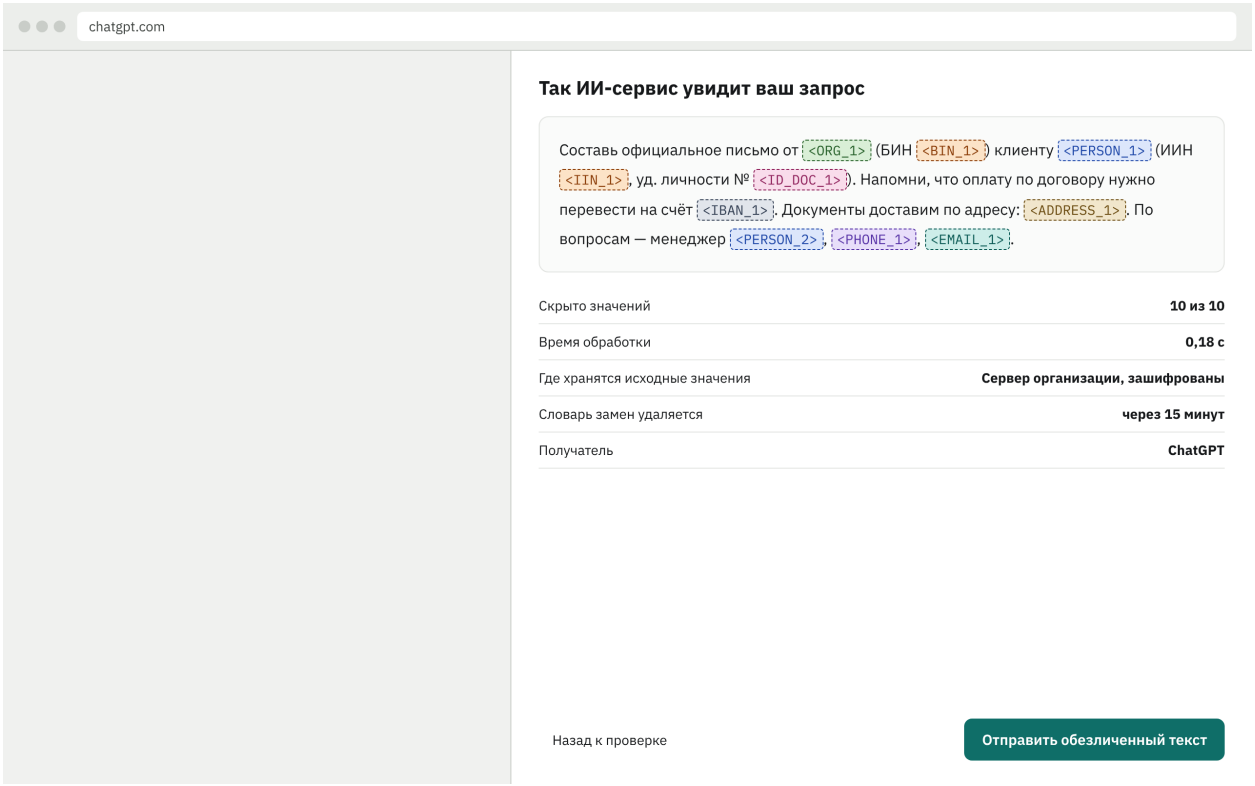


Figure 3: U3. Preview of the text that the AI service will receive and the processing time

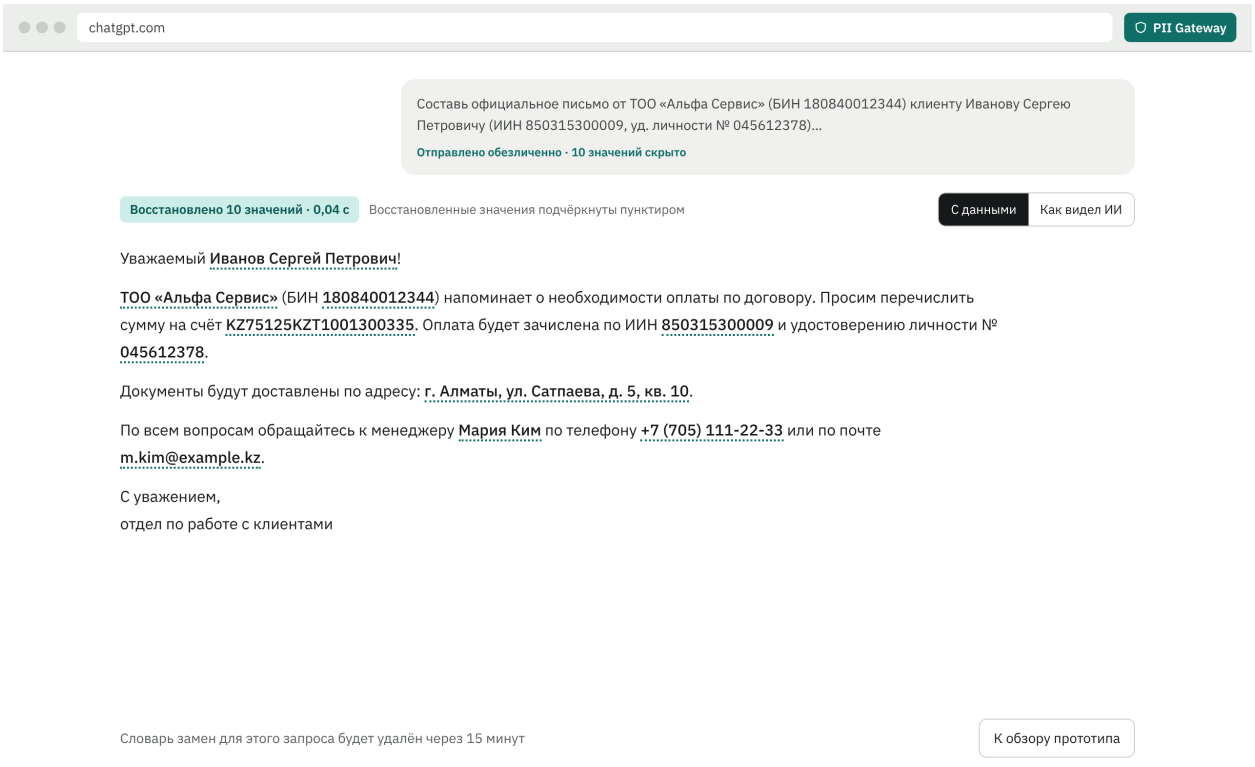


Figure 4: U4. Response with restored values: restored values are underlined, and the user can switch to the view "as the AI saw it"

5. Alternative Scenarios

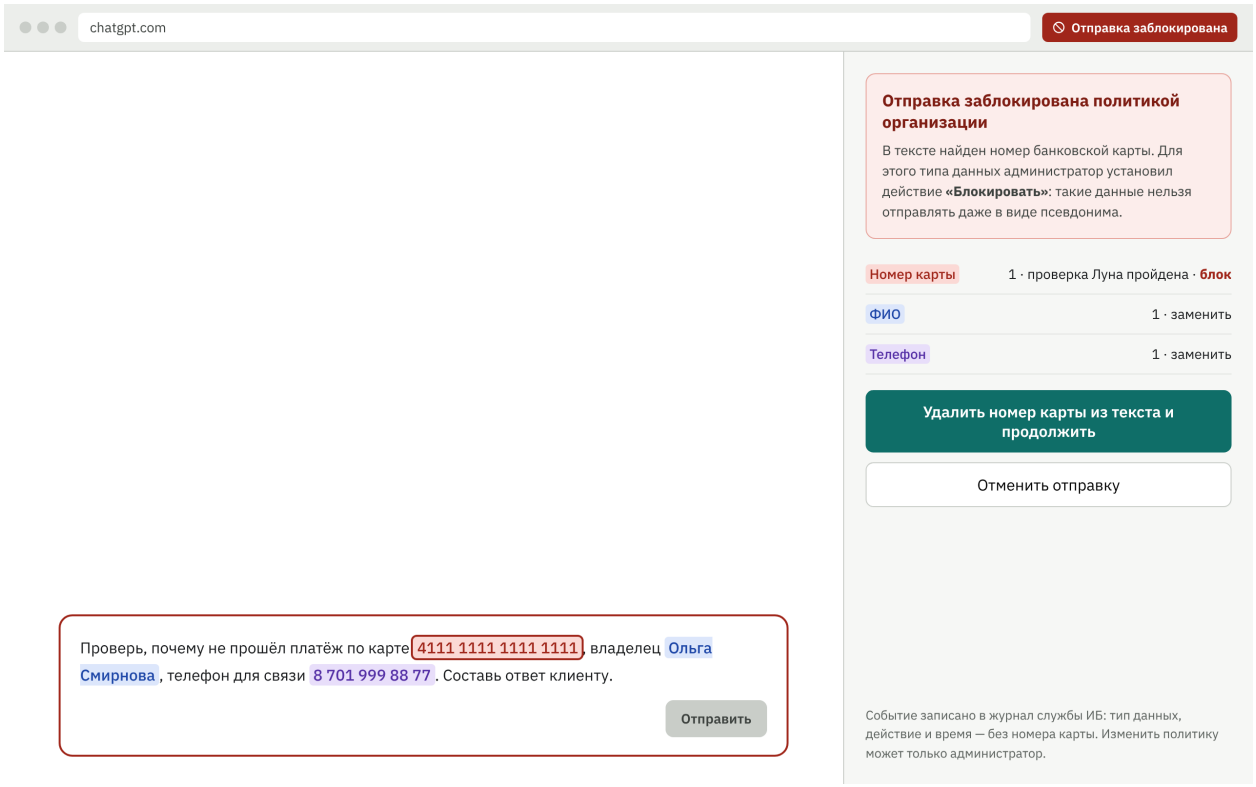


Figure 5: U5. Blocking by policy: the text contains a bank card number validated by the Luhn algorithm, for which the administrator set the action “block” (UC-3)

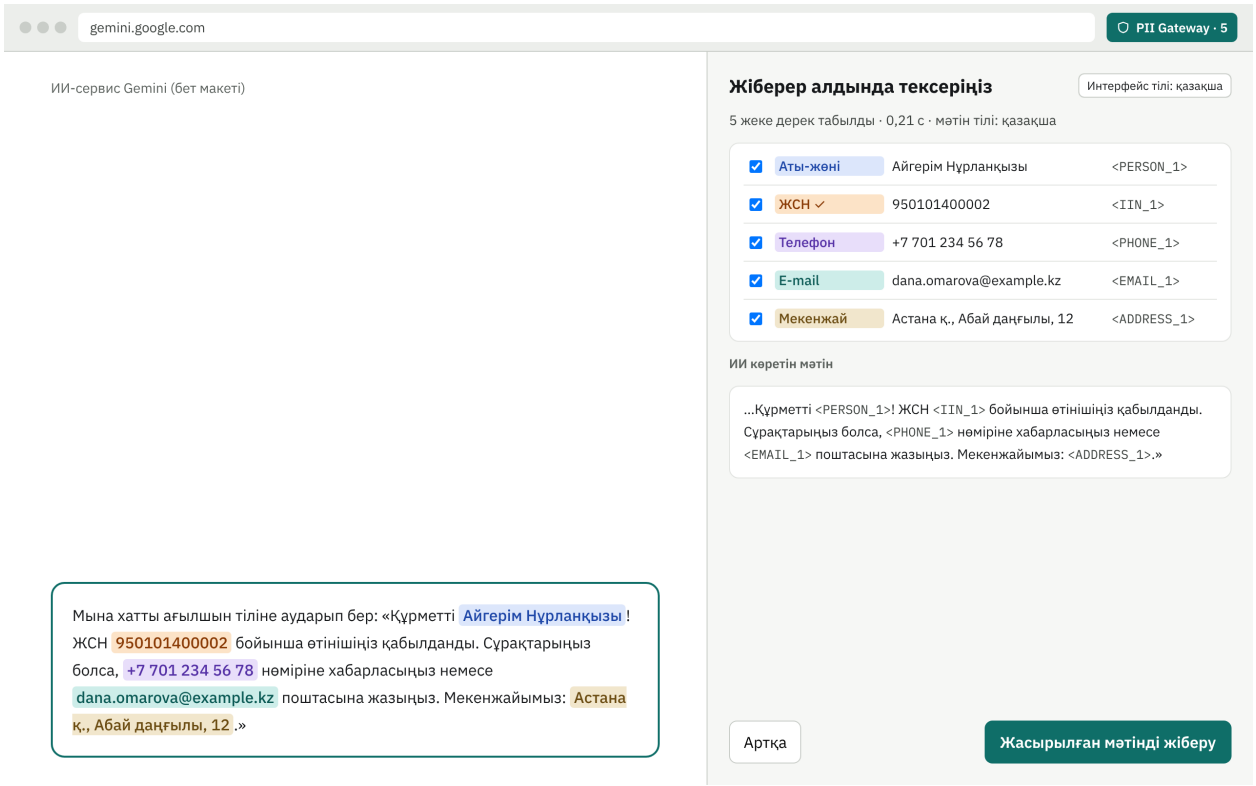


Figure 6: U6. Gemini with a Kazakh text and the Kazakh extension interface

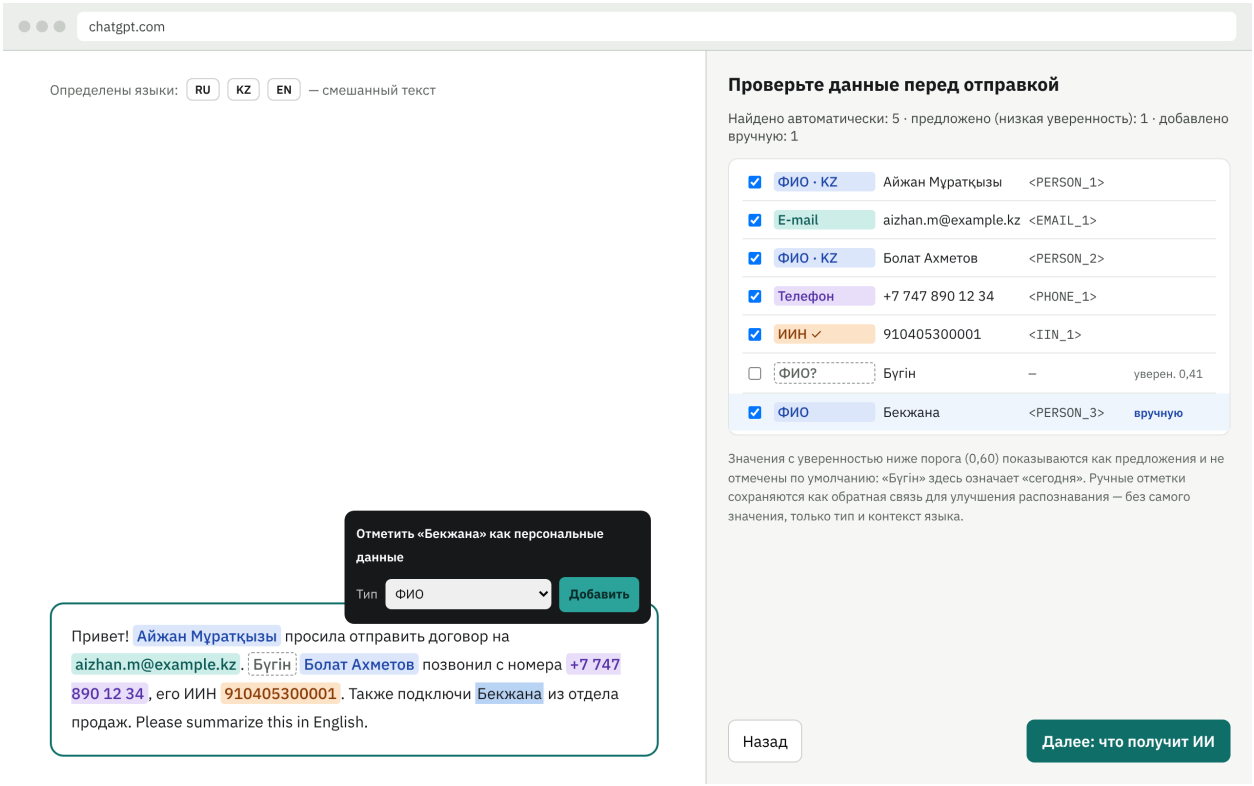


Figure 7: U7. Mixed Kazakh–Russian–English text: a low-confidence suggestion is not marked by default, and a missed name is added manually

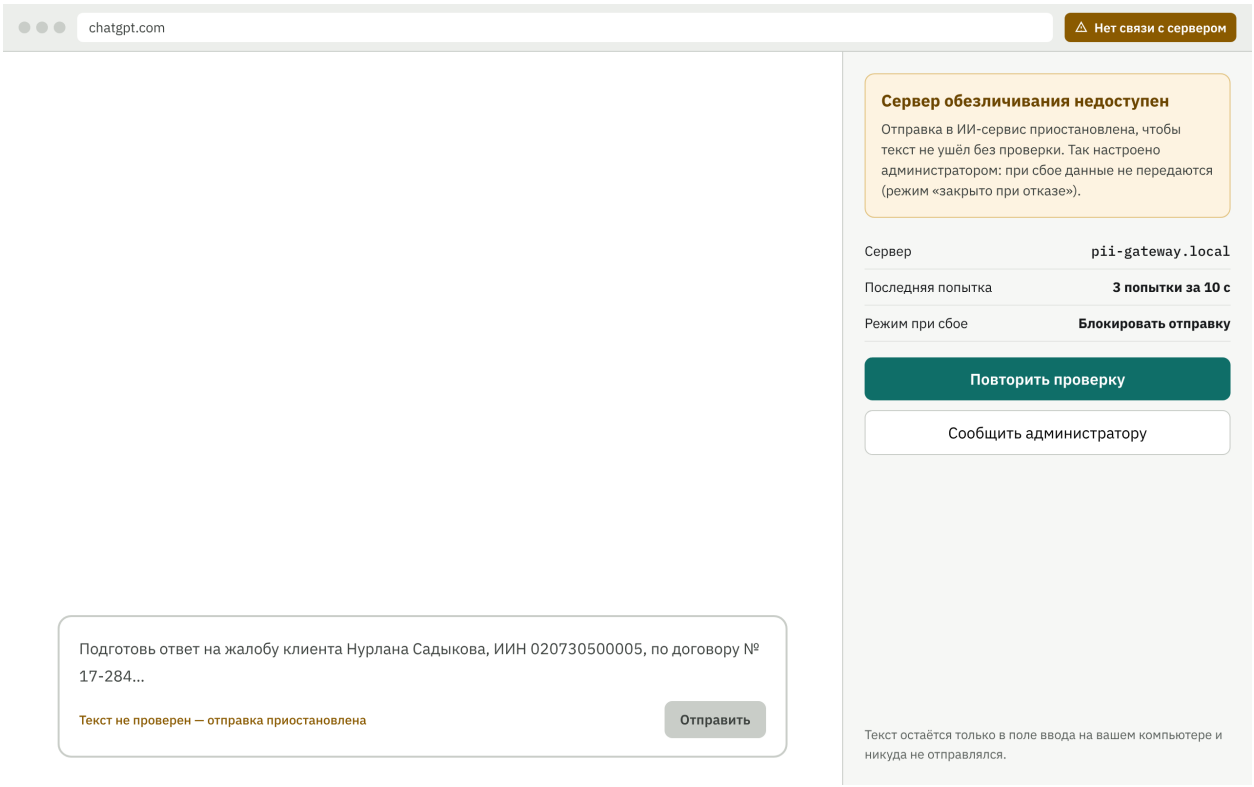


Figure 8: U8. Anonymization server unavailable: sending is suspended so that the text does not leave unchecked (fail-closed)

6. Extension Window

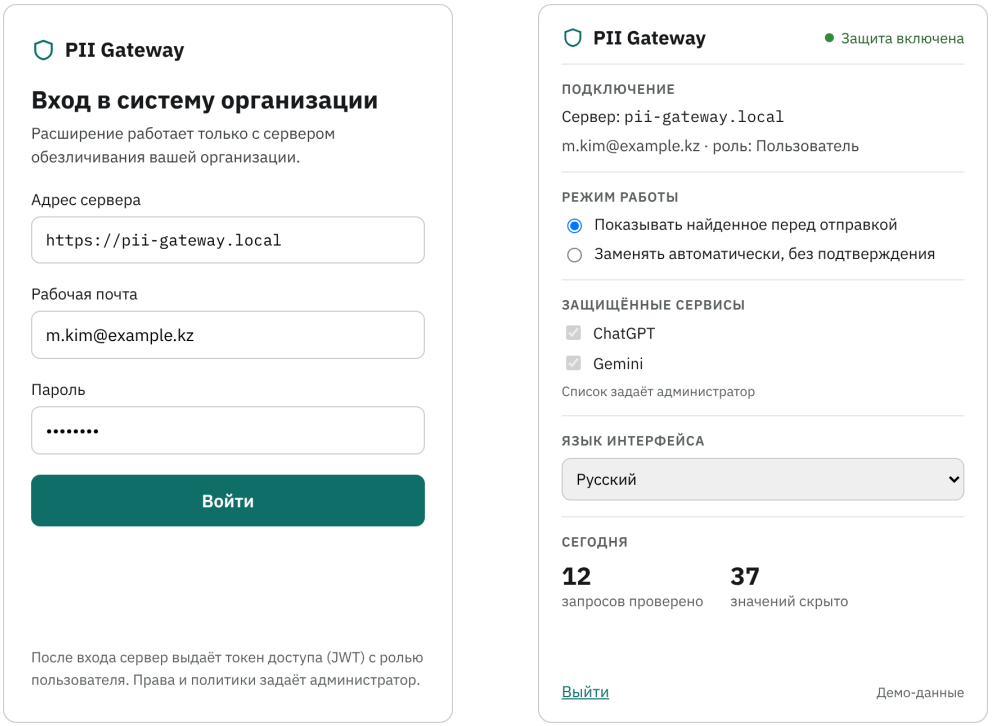


Figure 9: X2 (left): sign-in with an organization account; X1 (right): protection status, mode, protected services and interface language

7. Administration Panel

The administration panel is used by the information security service (use cases UC-5 and UC-6). Access is limited to the roles “Administrator” and “Auditor”.

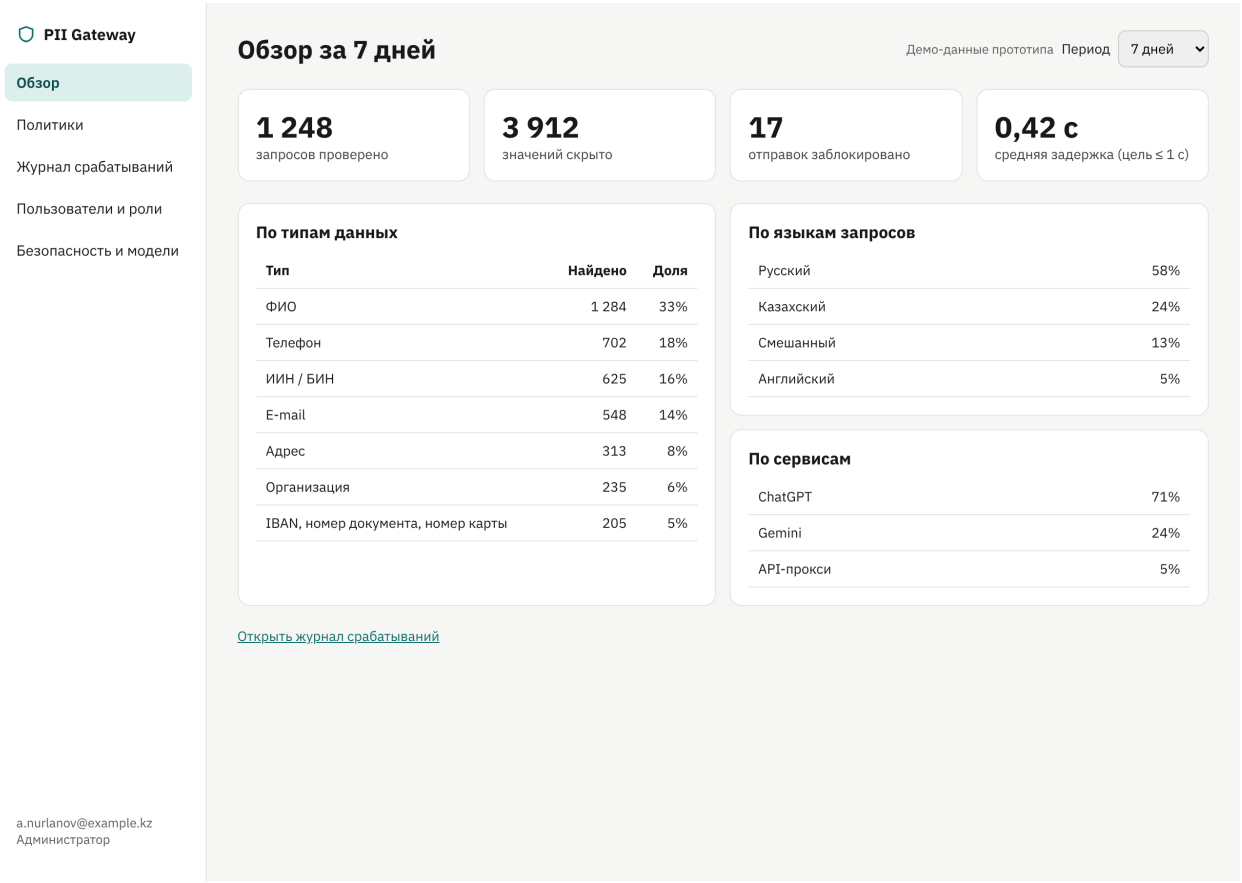


Figure 10: A2. Statistics: number of checks, hidden values, blocked requests, distribution by data type and average latency

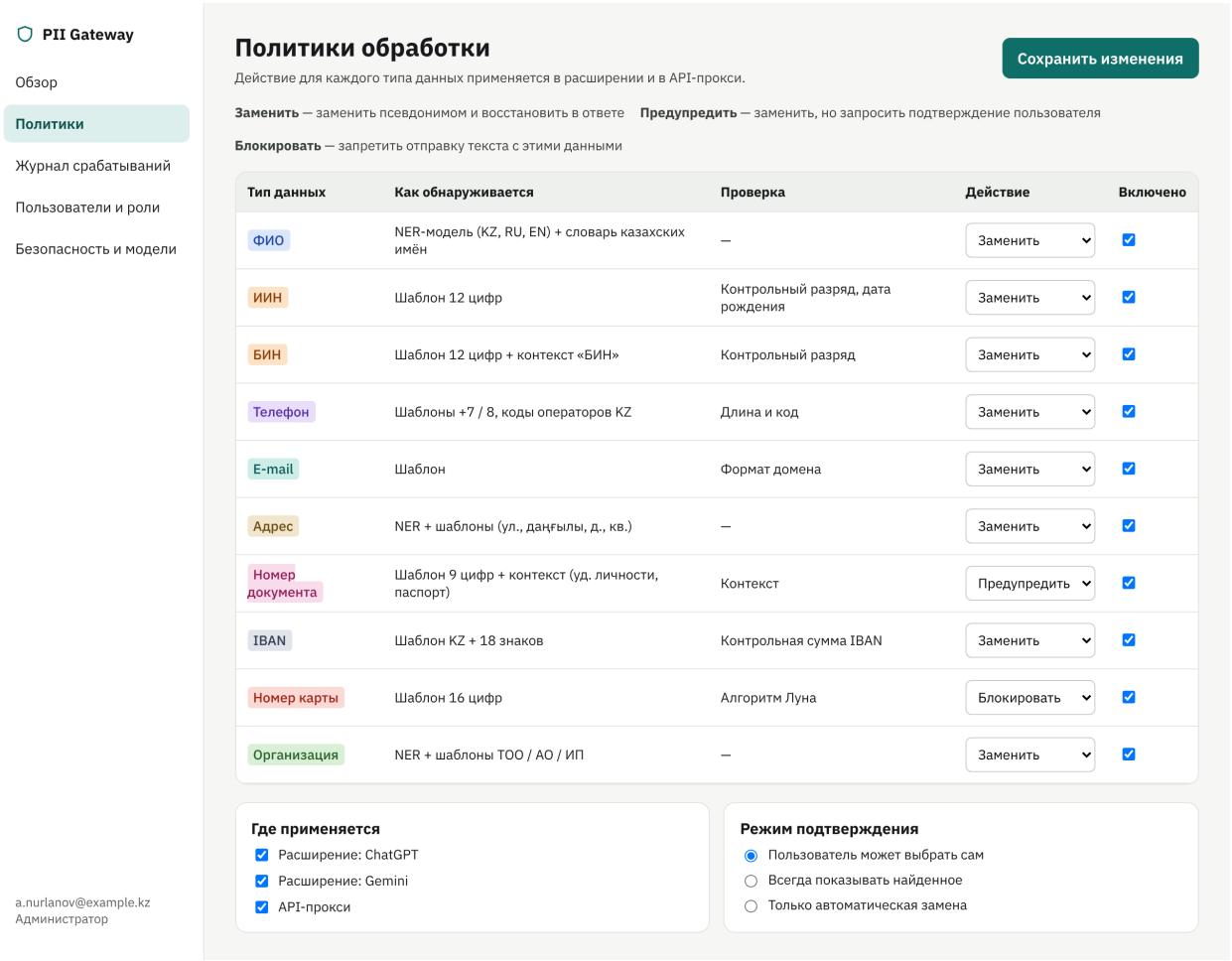


Figure 11: A3. Processing policies: detection method, validation and action for each data type

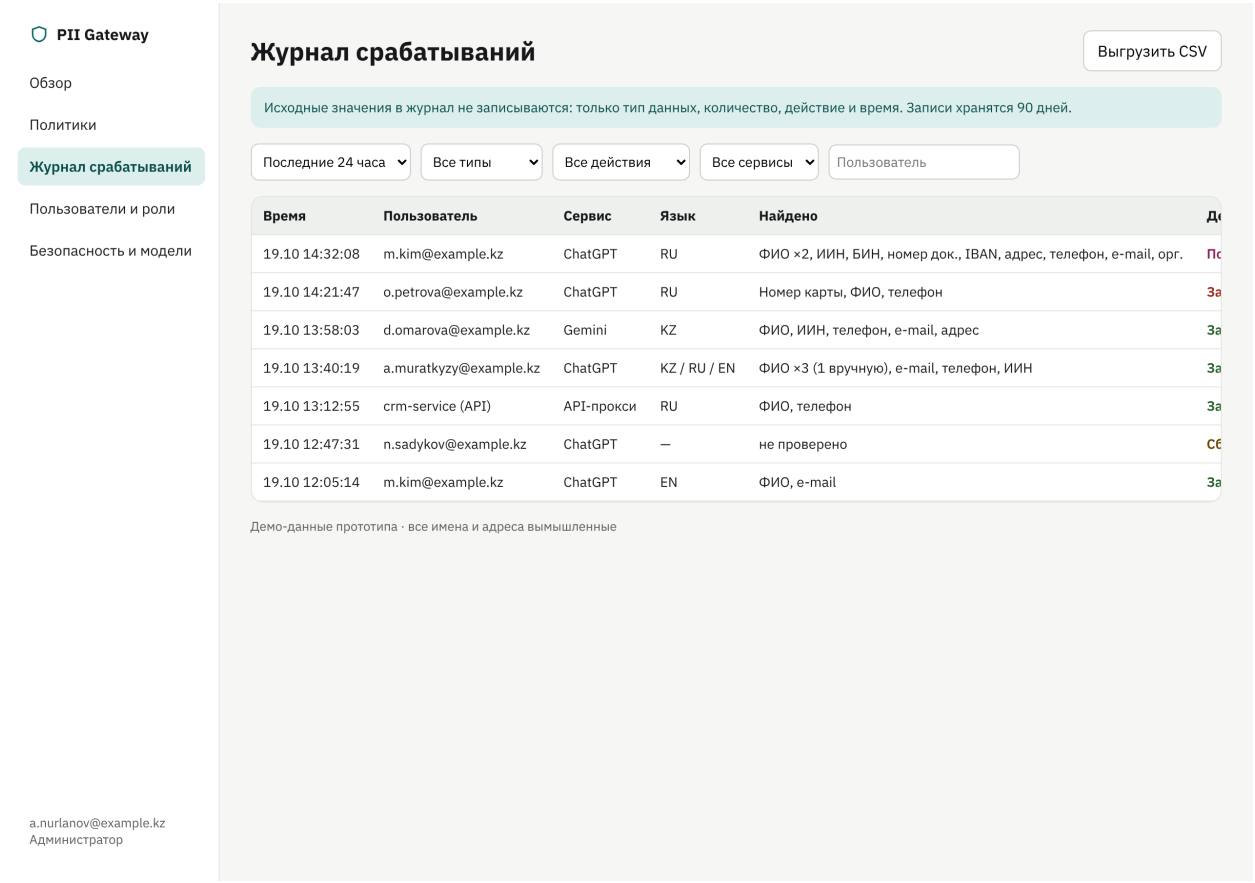


Figure 12: A4. Detection log: only the data type, number of values, action and time are recorded, without original values

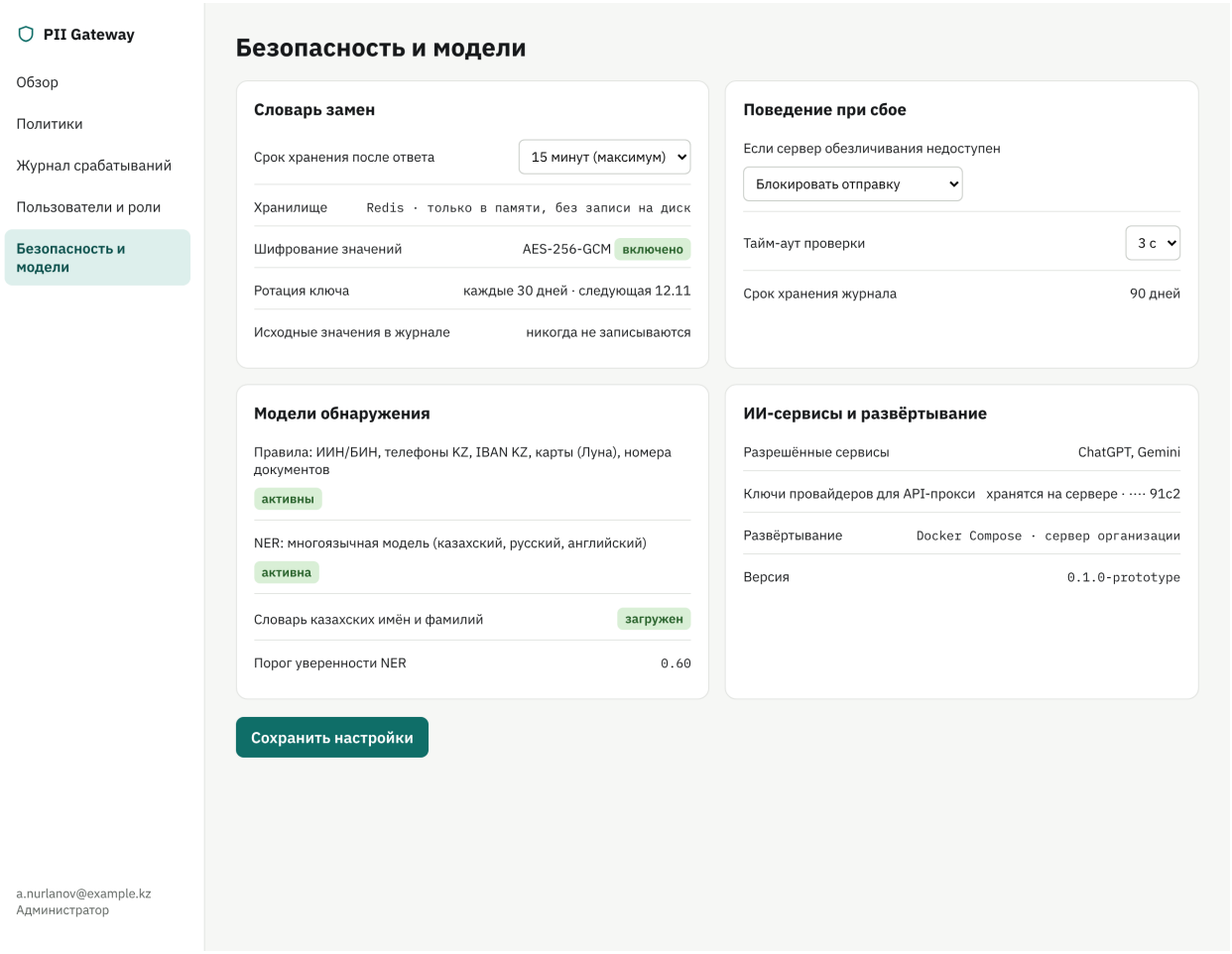


Figure 13: A6. Security and models: mapping retention (at most 15 minutes), encryption, behaviour on failure, log retention and status of detection models

Screens A1 (sign-in) and A5 (users and roles) are included in the repository and listed in Table 2.

8. Programming Interface

Screen P1 (Figure 14) shows the requests and responses of the anonymization endpoint and of the proxy compatible with the OpenAI chat interface, which are used by service accounts (UC-4).



Figure 14: P1. API examples: anonymization, proxy to the language model with a streamed response, and restoration

9. Screens and Requirements

Table 2 links each screen to the requirements it demonstrates. Every functional requirement FR-1–FR-8 and the user-facing non-functional requirements are covered by at least one screen.

Table 2: Screens and requirements

Screen	Name	What is shown	Requirements
U1	Text input	Highlighting of all data types, validated IIN and BIN	FR-1, FR-2, FR-5.1
U2	Review of detected data	Review list, unmarking, type change, manual marking, "warn" policy	FR-3.2, FR-5.2, FR-5.3, FR-6.1
U3	Preview	Text received by the AI service, processing time	FR-3.1, FR-5.4, NFR-1
U4	Restored response	Streamed response with restored values, view "as the AI saw it"	FR-4.1, FR-4.2, FR-4.4
U5	Blocking	Card number blocked by policy, reason shown	FR-2.2, FR-6.1, FR-6.2
U6	Kazakh text, Gemini	Kazakh interface, IIN in Kazakh text, second AI service	FR-1.1, FR-5.1, FR-5.7
U7	Mixed text	Detected languages, low-confidence suggestion, manual marking	FR-1.3, FR-5.2, FR-5.3
U8	Server unavailable	Sending suspended	NFR-4
X1	Extension window	Status, mode, protected services, interface language	FR-5.5, FR-5.7, FR-6.3
X2	Extension sign-in	Organization account, token	FR-5.6, NFR-2
A1	Panel sign-in	Access for administrators and auditors only	FR-7.4, NFR-2
A2	Statistics	Checks, hidden values, blocks, latency	FR-7.3, NFR-1
A3	Policies	Action per data type, where policies apply	FR-6.1, FR-6.3
A4	Log	Filters, CSV export, no original values	FR-7.1, FR-7.2
A5	Users and roles	Administrator, auditor, user, service account	FR-7.4
A6	Security and models	Retention, encryption, failure behaviour, NER threshold	FR-7.5, NFR-2, NFR-4
P1	API	Anonymization endpoint, OpenAI-compatible proxy, restoration	FR-4, FR-8

10. Usability Testing Plan

The prototype will be tested with 8–10 participants from the target groups (employees handling personal data and students) before the implementation of the extension user interface. Each participant performs three tasks: sending a request with personal data (U1–U4), handling a blocked request (U5) and adding missed data in a mixed-language text (U7). The following are measured: task completion rate, number of errors, time per task and the System Usability Scale (SUS) score. The target values are a task completion rate of at least 90% and a SUS score of at least 70 (NFR-5). The results will be used to revise the prototype; the detailed procedure is described in the document *Evaluation Plan*.

11. Limitations

The prototype is static: detection results and statistics on the screens are demonstration data and do not reflect the real quality of the system. The web pages of ChatGPT and Gemini are shown as simplified mock-ups, since their actual interfaces change frequently. The administration panel is shown only for desktop screens.